3. **Robustness of Linear Projections:**

Transformer layers mix representations linearly (plus nonlinear gating).

Because linear projections approximately preserve geometry (as JL guarantees for random ones), the model can safely compress and recombine information across layers without catastrophic loss of meaning.

4. **Distributed Semantics:**

The lemma supports the intuition that *semantic meaning doesn't need explicit symbolic coordinates*. Nuances can be represented as *distributed directions* in a continuous space — distances and angles encode relationships robustly under random or learned linear transformations.
